



WS2812B Series Product Upgrade Notice

Dear Worldsemi customers:

To continuously provide a better user experience, We have completed the technical iteration and upgrade of WS2812B series products. This upgrade maintains the communication protocol and package size exactly the same as the existing products, only optimizing the internal parameters. The specific improvements are as follows:

1.Ultra-Low Quiescent Power Consumption

IC standby current $\leq 1 \mu\text{A}$; Operating current reduced from 1 mA to 200 μA ; Achieving over 80% energy savings in basic power consumption.

2.Freely Interchangeable DIN/DOUT

Input/output pins can be swapped arbitrarily, simplifying PCB layout and enhancing the convenience of connecting more DIY products.

3.Constant-Current Drive at 3.3V

RGB channel voltage drop optimization, 3.3V power supply can achieve the same light emission consistency as 5V. When working with 3.3V power supply, energy can be saved by more than 30%.

4.Zero flicker when power-on

Completely eliminate random flickering upon power-on and enhance the overall visual experience during startup.

5.EMC upgrade again

Internal structure optimization, reduces radiated emissions by more than 6 dB, making it easier to pass radiated emission tests than before.

6.Control signal upgrade again

The control signal level is adjusted to 0.6 VDD.

It is ensured that the 3.3V signal level can fully control the device when 5V power supply,

After the upgrade, the model suffix will be uniformly V6 version. The unified naming rule is WS2812C-V6 (5050 package), WS2812C-2020-V6 (2020 package). – mixing old and new batches is seamless. Inventory switching starts today.

If you have any questions or need test samples, please feel free to contact us!

In addition, we also have a V7 version. Compared with V6, the cost increases by less than 20%, but the brightness can be increased by more than double. If you need a version with higher brightness, you can test and compare!

WORLDSEMI CO.,LIMITED

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